

Skills Summary

Radical Equations, Domains, and Decimals

Use this reference while completing your worksheet or when studying.

Skill 1 — The domain of a radical

Rule: $\sqrt{A}$ is defined exactly where $A \geq 0$. Set the radicand ≥ 0 and solve. If the radical sits in a *denominator*, use > 0 instead: the radicand may not be zero there.

Example: Give the domain of $f(x) = \sqrt{3x - 12}$.

Step 1. The only restriction a square root carries is that its radicand cannot be negative, so the domain question becomes an inequality.

Step 2. $3x - 12 \geq 0$ gives $3x \geq 12$, and dividing by the positive 3 keeps the direction.
 $\Rightarrow x \geq 4$

Try it: Give the domain of $g(x) = \sqrt{10 - 5x}$.

check: $x \geq 2$

Skill 2 — Isolate, then undo the radical

Method: strip everything off the radical first, then square both sides.

$$\sqrt{A} = B \implies A = B^2$$

Squaring before isolating leaves a cross term and makes the equation harder.

Example: Solve $\sqrt{x - 1} + 4 = 9$.

Step 1. Squaring is only worth doing once the root stands alone, so undo the +4 first.

Step 2. $\sqrt{x - 1} = 5$, so $x - 1 = 25$. Check: $\sqrt{25} + 4 = 9$.
 $\Rightarrow x = 26$

Try it: Solve $\sqrt{x + 3} - 2 = 3$.

check: $x = 22$

Skill 3 — Reject the extraneous candidate

Why it happens: squaring loses the sign, so the squared equation can have roots the original never had. Being inside the domain is *not* enough — every candidate must be substituted into the original equation. A candidate that makes the isolated radical equal a negative quantity always fails.

Example: Solve $\sqrt{x+12} = x$.

Step 1. The right side is a variable and could be negative, which is exactly when squaring manufactures a false root, so both candidates will need testing.

Step 2. $x+12 = x^2$ gives $x^2 - x - 12 = 0$, so $(x-4)(x+3) = 0$. Testing: $\sqrt{16} = 4$ works; $\sqrt{9} = 3$ is not -3 .

$\Rightarrow x = 4$ (the candidate -3 is extraneous)

Try it: Solve $\sqrt{x+20} = x$.

check: $x = 5$ (the candidate -4 is extraneous)

Skill 4 — Reporting a rounded decimal

Method: carry full precision through the working and round only at the very end, to the number of places asked for. Rounding an intermediate value and then squaring it magnifies the error.

Example: Solve $\sqrt{x-3} = 2.5$, giving x to two decimal places.

Step 1. The radical is already alone, so squaring releases x in one step, and the decimal only has to survive that one operation.

Step 2. $x-3 = 2.5^2 = 6.25$, so $x = 9.25$.

$\Rightarrow 9.25$

Try it: Solve $\sqrt{x+1} = 3.8$, giving x to two decimal places.

check: 13.44